College of Engineering – Department of Mechanical Engineering

CURRICULUM

Bachelor of Science in Naval Architecture and Marine Engineering

Academic Year 2025-2026

Reference CMOs: CMO No. 21 s. 2021, CMO No. 39 s. of 2021, CMO No. 27 s. of 2015, CMO No. 20 s. of 2013

Curriculum Description

The Naval Architecture and Marine Engineering (BSNAME) program equips students with the technical knowledge and practical skills needed to design, construct, and maintain marine vessels and offshore structures. It integrates mathematics, sciences, and engineering principles with specialized courses in ship design, marine power systems, and structural analysis, reinforced through laboratory work, field visits, and experiential learning.

The curriculum emphasizes leadership, ethics, and innovation, preparing graduates to apply engineering solutions responsibly in global, environmental, and societal contexts. Modern pedagogies such as project-based learning, technology-enhanced instruction, and the CDIO framework nurture collaboration, creativity, and problem-solving. Integration of Sustainable Development Goals, gender equality, and peace studies ensures students develop a holistic perspective aligned with contemporary challenges in the maritime industry.

Graduates are envisioned as specialists, leaders, and lifelong learners who contribute to sustainable maritime development and national progress. Career paths include naval architect, marine engineer, shipyard manager, classification surveyor, consultant, researcher, and academician, with allied opportunities in mechanical, civil, marine, and ocean engineering. The program fosters professionals who embody excellence, integrity, resilience, and service—ready to advance the maritime sector responsibly.

Program Education Objectives

The naval architecture and marine engineering alumni three to five years after graduation shall:
PEO 1: Practiced as specialist in solving complex naval architecture and marine engineering problems leading to improvements and innovations, while taking into consideration the environmental, social, and economical requirements.
PEO 2: Assume leadership position in multidisciplinary workforce in industry, academe, government, or private sector both local and global with consideration to social and ethical responsibilities.
PEO 3: Engage in lifelong learning through further studies, research, certifications, promotions, and other personal and professional development activities.

Student Outcomes

The following skills, knowledge, and values are expected to be attained by students as they progress through the program:
SO1 Discipline Knowledge. Ability to apply mathematics, sciences and principles of engineering to solve complex naval architecture and marine engineering problems;

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SO2 Investigation. Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions;

SO3 Design/Development of Solutions. Design solution, system, components, processes, exhibiting improvements/innovations, that meet specified needs with appropriate consideration for public health and safety, cultural, societal, economical, ethical, environmental and sustainability issues.

SO4 Leadership and Teamwork. Function effectively as a member of a leader on a diverse team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

SO5 Problem Analysis. Identify, formulate, and solve complex naval architecture and marine engineering problems by applying principles of engineering, science, and mathematics;

SO6 Ethics and Professionalism. Apply ethical principles and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, environmental, and societal contexts.

SO7 Communication. Communicate effectively on complex naval architecture and marine engineering activities with the community, and the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions;

SO8 Environment and Sustainability. Recognize the impact of professional engineering solutions in societal, global, and environmental contexts and demonstrate knowledge of and need for sustainable development;

SO9 Lifelong Learning. Recognize the need for, and ability to engage in independent and life-long learning in the broadest context of technological change.

SO10 The Engineer and Society. Apply reasoning based on contextual knowledge to assess societal, health, safety, legal, cultural, contemporary issues, and the consequent responsibilities relevant to professional engineering practices.

SO11 Modern Tool Usage. Apply appropriate techniques, skills, and modern engineering and IT tools to complex mechanical engineering activities;

SO12 Project Management and Finance. Demonstrate knowledge and understanding of engineering management and financial principles as member or a leader of a team to manage projects in multidisciplinary settings and identify opportunities of entrepreneurship.

SO13 Social and National Responsibility. Apply acquired engineering knowledge and skills in addressing community problems that contributes to national development.

CURRICULUM COMPONENTS

Classification / Field / Course	Number of Hours per Week		Credit Units
	Lec	Lab	
I. TECHNICAL COURSES			
A. Mathematics			
Differential Calculus	4	0	4
Integral Calculus	4	0	4
Differential Equations	3	0	3
Subtotal	**11**	**0**	**11**
B. Natural/ Physical Sciences			
Chemistry for Engineers	3	3	4
Physics for Engineers	3	3	4
Subtotal	**6**	**6**	**8**
C. Basic Engineering Sciences			
Engineering Economics	3	0	3
Technopreneurship	3	0	3

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Basic Occupational Safety and Health	3	0	3
Environmental Science and Engineering	3	0	3
Statics of Rigid Bodies	3	0	3
Dynamics of Rigid Bodies	3	0	3
Methods of Research	3	0	3
Mechanics of Deformable Bodies	3	0	3
Computer Programming	0	3	1
Subtotal	**24**	**3**	**25**
D. Allied Courses			
Basic Electrical Engineering	2	3	3
Basic Electronics	2	3	3
DC and AC Machinery	3	3	4
Subtotal	**7**	**9**	**10**
E. Fundamental NAME Courses			
Thermodynamics 1	3	0	3
Workshop Theory and Practice for NAME	2	3	3
Thermodynamics 2	3	0	3
Fluid Mechanics	3	0	3
Machine Elements	3	0	3
Subtotal	**14**	**3**	**15**
F. Professional NAME Courses			
Introduction to Naval Architecture and Marine Engineering	3	3	4
Ship Drafting 1	0	6	2
Ship Drafting 2	0	6	2
Ship Modeling	2	3	3
Ship Hydrostatics	3	0	3
Ship Resistance	3	3	4
Intact Stability	3	3	4
Damage Stability	3	6	5
Ship Structural Analysis	3	6	5
Ship Structural Design	3	3	4
Motion in Waves	2	3	3
Marine Engineering 1	3	3	4
Marine Engineering 2	3	3	4
Marine Auxiliary Systems	3	0	3
Marine Power Plant Engineering	2	3	3
Ship Propulsion	3	3	4
Maritime Laws, PMMRR, and Code of Ethics	3	0	3
Ship Building Processes	3	6	5
Shipyard Processes	2	3	3
Shipyard Management	3	0	3
Contracts and Specifications	3	0	3
Marine Inspection and Surveys	3	0	3
Ship Design 1	2	9	5
Ship Design 2	2	9	5
Naval Architecture and Marine Engineering Practice and Comprehensive Exam	0	6	2
Material Science and Engineering	3	0	3

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Subtotal	**63**	**87**	**92**
G. On-the Job training			
OJT / Internship	400 Hrs		5
Subtotal	**0**	**0**	**5**
TOTAL TECHNICAL COURSES	**125**	**108**	**166**
II. NON-TECHNICAL COURSES			
A. General Education Courses			
Understanding the Self	3	0	3
Mathematics in the Modern World	3	0	3
The Contemporary World	3	0	3
Readings in Philippine History	3	0	3
Purposive Communication	3	0	3
Ethics	3	0	3
Art Appreciation	3	0	3
Science, Technology and Society	3	0	3
Subtotal	**24**	**0**	**24**
B. General Education Electives and Mandated Courses			
Linear Algebra for AI	3	0	3
Machine Learning and Neural Networks	3	0	3
Probability and Statistics for AI	3	0	3
Life and Works of Rizal	3	0	3
Subtotal	**12**	**0**	**12**
C. Physical Activities Toward Health and Fitness (PATHFit)			
Movement Competency Training	2	0	2
Exercise-based Fitness Activities	2	0	2
Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 1	2	0	2
Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 2	2	0	2
Subtotal	**8**	**0**	**8**
D. National Service Training Program			
National Service Training Program 1	3	0	3
National Service Training Program 2	3	0	3
Subtotal	**6**	**0**	**6**
TOTAL NON-TECHNICAL COURSES	**40**	**0**	**40**
GRAND TOTAL	**165**	**108**	**216**

SUMMARY

Courses	Number of Units
I. Technical Courses	
A. Mathematics	11
B. Natural/Physical Science	8
C. Basic Engineering Sciences	25
D. Allied Courses	10
E. Fundamental Naval Architecture and Marine Engineering Courses	**15**
F. Professional Engineering Courses	**92**
G. On-the-Job training	**5**

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II. Non-Technical Courses	
A. Required General Education	24
B. General Education Elective and Mandated Courses	12
C. Physical Activities Toward Health and Fitness (PATHFit)	8
D. National Service Training Program	6
Grand Total	**216**

CURRICULUM STRUCTURE

BACHELOR OF SCIENCE IN NAVAL ARCHITECTURE AND MARINE ENGINEERING

Reference CMOs: CMO No. 21 s. 2021, CMO No. 39 s. 2021,
CMO No. 27 s. 2015, CMO No. 20 s. 2013

FIRST YEAR							
FIRST SEMESTER							
Course Code	Course Title	Credit Units	Lec	lab	Prerequisite(s)	Co-requisite(s)	Category
MATH 401	Differential Calculus	4	4	0			TC-A
NAME 101	Introduction to Naval Architecture and Marine Engineering	4	3	3			TC-F
NAME 102	Ship Drafting 1	2	0	6			TC-F
Ged 101	Understanding the Self	3	3	0			NTC-A
Ged 102	Mathematics in the Modern World	3	3	0			NTC-A
CHEM 101	Chemistry for Engineers	4	3	3			TC-B
Ged 106	Purposive Communication	3	3	0			NTC-A
PATHFit 1	Movement Competency Training	2	2	0			NTC-C
NSTP 111	National Service Training Program 1	3	3	0			NTC-D
	Total	**28**	**24**	**12**			

FIRST YEAR							
SECOND SEMESTER							
Course Code	Course Title	Credit Units	Lec	lab	Prerequisite(s)	Co-requisite(s)	Category
MATH 102	Integral Calculus	4	4	0	MATH 101		TC-A
PHYS 101	Physics for Engineers	4	3	3	MATH 101	MATH 102	TC-B
GEd 104	The Contemporary World	3	3	0			NTC-A
GEd 108	Art Appreciation	3	3	0			NTC-A
GEd 109	Science, Technology and Society	3	3	0			NTC-A
CpE 101	Computer Programming	1	0	3			TC-D
PATHFit2	Exercise Based Fitness Activities	2	2	0	PATHFit 1		NTC-C
NSTP 121	National Service Training Program 2	3	3	0			NTC-D
GEd 105	Readings in the Philippine	3	3	0			NTC-A

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	History						
NAME 103	Ship Drafting 2	2	0	6	NAME 102		TC-F
	Total	**28**	**24**	**12**			
FIRST YEAR							
MIDTERM							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
EE 205	Basic Electrical Engineering	3	2	3	PHY 102, MATH 102		TC-D
GEd 103	Life and Works of Rizal	3	3	0			NTC-B
GEd 107	Ethics	3	3	0			NTC-A
	TOTAL	**9**	**8**	**3**			

SECOND YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
AI 101	Linear Algebra	3	3	0	MATH 102		NTC-B
ENGG 107	Statics of Rigid Bodies	3	3	0	PHY 102, MATH 102		TC-C
MATH 101	Differential Equation	3	3	0	MATH 102		TC-A
ME 202	Thermodynamics 1	3	3	0	PHY 102, MATH 102		TC-E
NAME 201	Ship Modeling	3	2	3	NAME 202		TC-F
NAME 202	Ship Hydrostatics	3	3	0	NAME 101		TC-F
NAME 204	Workshop Theory and Practice for NAME	3	2	3			TC-E
NAME 205	Marine Engineering 1	4	3	3			TC-F
PATHFit 3	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 1	2	2	0	PATHFit 2		NTC-C
	Total	**27**	**24**	**9**			

SECOND YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
ENGG 123	Dynamics of Rigid Bodies	3	3	0	ENGG 107		TC-C
ENGG 118	Mechanics of Deformable Bodies	3	3	0	ENGG 107		TC-C
NAME 203	Ship Resistance	4	3	3	NAME 203		TC-F
ME 205	Thermodynamics 2	3	3	0	ME 202		TC-E
ENGG 112	Fluid Mechanics	3	3	0	ME 202		TC-E
NAME 207	Ship Building Processes	5	3	6	NAME 101		TC-F
NAME 206	Marine Engineering 2	4	3	3	NAME 205		TC-F
PATHFit 4	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 2	2	2	0	PATHFit 3		NTC-C
	Total	**27**	**23**	**12**			

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SECOND YEAR							
MIDTERM							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
AI 102	Probability and Statistics for AI	3	3	0	MATH 102		NTC-B
ME 304	Machine Elements	3	2	3	ENGG 123		TC-C
ENGG 118	Material Science and Engineering	3	3	0	PHY 102		TC-F
	Total	**9**	**8**	**3**			

THIRD YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
NAME 305	Marine Power Plant Engineering	3	2	3	NAME 206, ME 205	NAME 304	TC-F
ENGG 116	Methods of Research	3	3	0	MATH 103		TC-C
EE 211	AC and DC Machinery	4	3	3	EE 205		TC-D
AI 103	Machine Learning and Neural Networks	3	2	3	AI 102		NTC-B
NAME 301	Intact Stability	4	3	3	NAME 203		TC-F
NAME 304	Marine Auxiliary Systems	3	3	0	NAME 206	NAME 305	TC-F
ECE 406	Basic Electronics	3	2	3	EE 419		TC-D
	Total	**23**	**18**	**15**			

THIRD YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
ENGG 113	Environmental Science and Engineering	3	3	0	CHEM 101		TC-C
NAME 302	Damage Stability	5	3	6	NAME 301		TC-F
NAME 303	Motion in Waves	3	2	3	ENGG 112, NAME 203, NAME 301	NAME 307	TC-F
NAME 306	Ship Propulsion	4	3	3	NAME 206, NAME 203		TC-F
NAME 307	Ship Design 1	5	2	9	3rd Year Standing	NAME 303	TC-F
	Total	**20**	**13**	**21**			

THIRD YEAR							
MIDTERM							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
ENGG 424	On-the-Job Training (400 Hours)	5	4oo hrs		NAME 301		TC-F
ENGG 114	Basic Occupational Safety and Health	3	3	0			TC-C
	Total	**8**	**13**	**21**			

FOURTH YEAR							
FIRST SEMESTER							
Course	**Course Title**	**Credit**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(**	**Category**

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Code		Units				s)	
NAME 403	Maritime Laws, PMMRR and Code of Ethics	3	3	0	GEd 107		TC-F
ENGG 105	Technopreneurship	3	3	0	3rd Year Standing		TC-C
ENGG 104	Engineering Economics	3	3	0	3rd Year Standing		TC-C
NAME 406	Contracts and Specifications	3	3	0	GEd 107		TC-F
NAME 401	Ship Structural Analysis	5	3	6	NAME 202, ENGG 118	NAME 402	TC-F
NAME 402	Ship Structural Design	4	3	3	NAME 202, ENGG 118	NAME 401	TC-F
	Total	**21**	**16**	**15**			

FOURTH YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
NAME 404	Shipyard Processes	3	2	3	NAME 207		TC-F
NAME 405	Shipyard Management	3	3	0	NAME 207		TC-F
NAME 407	Marine Inspection and Surveys	3	2	3			TC-F
NAME 408	Ship Design 2	5	2	9	NAME 307		TC-F
NAME 409	Naval Architecture and Marine Engineering Practice and Comprehensive Exam	2	0	6	4th Year Standing		TC-F
	Total	**16**	**9**	**21**			

TOTAL UNITS: 216